For the anomalous acceleration of the universe's expansion and the dark energy that presumably causes it, I recommend the last chapter of *The Cosmic Cocktail* (Freeze 2014) and also *The 4% Universe: Dark Matter, Dark Energy, and the Race to Discover the Rest of Reality* (Panek 2011).

Chapter 25. The Professor's Equation

The ideas that Newton's gravitational constant *G* might change from place to place and time to time, and might be controlled by some sort of nongravitational field, were hot topics in the Princeton University physics department when I was a PhD student there in the early 1960s. These ideas had been proposed by Princeton's Professor Robert H. Dicke and his graduate student Carl Brans in connection with their "Brans-Dicke theory of gravity" (Chapter 8 of *Was Einstein Right?* [Will 1993]), an interesting alternative to Einstein's general relativity. For a brief personal memoir about this, see "Varying Newton's Constant: A Personal History of Scalar-Tensor Theories" in *Einstein Online* (Brans 2010). The Brans-Dicke theory has motivated a number of experiments that searched for varying *G*, but no convincing variations were ever found; see, for example, Chapter 9 of *Was Einstein Right?* (Will 1993). These ideas and experiments motivated my interpretation of some of *Interstellar*'s gravitational anomalies and how to control them: bulk fields control the strength of *G* and make it vary.

The Professor's equation, shown on his blackboard in Figure 25.6, builds on these ideas. It also incorporates Einstein's relativistic laws (general relativity), extended into the bulk's fifth dimension, which are laid out in a technical review article by Roy Maartens and Koyama Kazuya (Maartens and Kazuya 2010), and it incorporates a branch of mathematics called the "calculus of variations"; see, for example,